🏆 The Definitive MAANG System Design Master Canon 0️⃣Core Foundations (Non-Negotiable) Computer Science Fundamentals • Time & space complexity (incl. amortized analysis) • Memory hierarchy (CPU cache → RAM → Disk/SSD) • CPU-bound vs I/O-bound workloads • Cache locality & data access patterns Operating Systems • Processes vs threads • Context switching • Locks, mutexes, semaphores • Deadlocks & starvation • Virtual memory basics Networking • TCP vs UDP • HTTP/1.1 vs HTTP/2 vs HTTP/3 • DNS resolution • TLS handshake (conceptual) • Latency vs bandwidth • Connection pooling 1️⃣ High-Level Design (HLD) 1.1 Requirements & Estimation • Functional vs non-functional requirements • Latency targets • Throughput targets • Availability (e.g., 99.9 vs 99.99) • Back-of-the-envelope calculations: (QPS, Storage estimation, Memory feasibility, Bandwidth estimation) 1.2 Data Management & Storage • Relational Databases: B-Trees (why optimized for disk), Indexing trade-offs, Write-Ahead Logging (WAL), ACID properties, Isolation levels, Read replicas, Master-slave vs multi-writer • NoSQL Databases: Key-value stores, Document databases, Wide-column stores, When NOT to use NoSQL • Data Modeling: Normalization vs denormalization, Read-heavy vs write-heavy systems, Hot partitions, Secondary indexes 1.3 Consistency & Distributed Coordination • CAP theorem (deep intuition) • PACELC theorem • Strong vs eventual consistency • Quorum (R/W/N) • Leader election • Distributed locks • Saga pattern • 2-Phase Commit (2PC) 1.4 Scaling & Partitioning • Vertical vs horizontal scaling • Stateless vs stateful services • Sharding strategies: (Hash-based, Range-based, Geo-based) • Rebalancing & resharding • Hot key problem 1.5 Communication Models & Real-Time Systems ⭐ • Request–Response: REST APIs, gRPC, GraphQL, Idempotency, Pagination, Versioning • Client-Pull Models: Polling (fixed interval), Long polling • Server-Push Models: WebSockets (full-duplex), Server-Sent Events (SSE) • Choosing Between Them: Chat → WebSockets, Live feed → SSE, Periodic dashboard → Polling, Internal service calls → gRPC, Public APIs → REST 1.6 Caching • Cache layers (CDN, service, DB) • LRU, LFU eviction • Write-through vs write-back • TTL strategies • Cache invalidation • Cache stampede • Hot keys 1.7 Asynchronous & Event-Driven Systems • Message queues vs event streams • At-most-once / at-least-once / exactly-once • Idempotent consumers • Dead-letter queues • Backpressure handling • Event-driven architecture 1.8 Reliability & Fault Tolerance • Replication • Redundancy • Graceful degradation • Retries with exponential backoff • Circuit breakers • Bulkheads • Cell-based architecture • Blast radius containment 1.9 Observability & Operations • Metrics, logs, distributed tracing • Liveness vs readiness probes • SLIs, SLOs, SLAs • Canary deployments • Blue-green deployments 1.10 Security • Authentication vs authorization • OAuth / JWT (conceptual) • Rate limiting • DDoS protection • Encryption in transit & at rest • Secrets management 2️⃣Low-Level Design (LLD) 2.1 Object-Oriented Design • SOLID principles (applied) • Encapsulation, abstraction • Composition vs inheritance • Immutability • Cohesion & coupling 2.2 High-Signal Design Patterns • Creational: Factory, Builder, Singleton (and why dangerous) • Structural: Adapter, Decorator, Facade, Proxy • Behavioral: Strategy, Observer, State, Command 2.3 UML & Modeling • Class diagrams, Sequence diagrams, State diagrams, Entity-relationship modeling 2.4 Concurrency & Thread Safety • Thread safety principles • Optimistic vs pessimistic locking • Atomic operations (CAS intuition) • Thread pools • Producer–consumer pattern 2.5 Performance Engineering • Profiling, Memory leaks, GC basics • Object lifecycle, Lazy loading, Batching 3️⃣End-to-End System Design Practice • URL shortener • News feed • Chat system (WebSockets at scale) • Notification system • Payment workflow • File storage system • For each: APIs, Data model, Caching, Scaling, Failures, Trade-offs 4️⃣Architect-Level Thinking (The Real Differentiator) • Latency vs consistency • Cost vs reliability • Simplicity vs extensibility • Strong vs eventual consistency • Build vs buy decisions • Cost awareness • Over-engineering detection • Failure-first design • Rollbacks & recovery 5️⃣Interview Execution Framework • HLD Flow: Clarify requirements, Define assumptions, Estimate scale, Define APIs, Design data model, High-level architecture, Bottlenecks, Failure handling, Trade-offs • LLD Flow: Clarify use cases, Identify entities, Design classes, Define relationships, Handle concurrency, Ensure extensibility 🌟 6️⃣Elite MAANG Appendices • 6.1 Advanced Distributed Data & Engines: LSM-Trees vs B-Trees, SSTables & Memtables, Compaction strategies, Online schema changes at scale • 6.2 Time & Ordering in Distributed Systems: Clock drift, Lamport clocks, Vector clocks, TrueTime • 6.3 Advanced Networking & Infrastructure: TCP BBR vs Cubic, Anycast & BGP, Edge computing, Service mesh concepts, Sidecar pattern, Control plane vs data plane • 6.4 Probabilistic & Specialized Data Structures: Bloom filters, HyperLogLog, Quad-trees, Geohashing • 6.5 High-Scale Data Engineering: Batch vs stream processing, Change Data Capture (CDC), Backpressure & load shedding 🆕 7️⃣Staff / Principal-Level Additions • 7.1 Edge, Gateway & Traffic Management: API gateways, Global load balancing, Request shaping, Throttling, WAF concepts, Edge authentication, Multi-layer rate limiting • 7.2 Multi-Region & Disaster Recovery: Active-active vs active-passive, RPO & RTO, Cross-region replication lag, Geo-fencing & data residency, Region failover strategies, Split-brain avoidance • 7.3 Data Lifecycle & Storage Economics: Hot / warm / cold storage tiers, Tiered storage policies, Archival pipelines, Data retention strategies, GDPR-style deletions, Cost-aware storage design • 7.4 Deployment, Migration & Evolution: Infrastructure as Code mindset, Immutable infrastructure, Feature flags, Dark launches, Schema versioning, Online migrations, Roll-forward vs rollback • 7.5 Rate Limiting Algorithms (Deep Dive): Token bucket, Leaky bucket, Fixed window, Sliding window log, Sliding window counter, Distributed rate limiting • 7.6 Advanced Security & Compliance: PII segregation, Encryption key rotation, Audit logging, RBAC vs ABAC, Least privilege enforcement, Zero-trust principles • 7.7 Advanced Scaling & Migration Patterns: Shadow traffic, Dual writes, Read/write splitting, Strangler Fig pattern, Monolith → microservices evolution, Contract testing • 7.8 Decision Narration & Staff-Level Communication: Assumption declaration, Multiple-option framing, Explicit trade-off comparison, Justified decision making, Rejected-alternative explanation, Business impact alignment